TASK -1

```
mysql> USE operations_db;
Database changed
mysql> SELECT DATABASE();
+---------------+
| DATABASE()    |
+---------------+
| operations_db |
+---------------+
1 row in set (0.00 sec)

mysql> CREATE TABLE customer (
    ->     cus_name VARCHAR(20),
    ->     reg_id INT,
    ->     contact BIGINT
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> ALTER TABLE customer ADD location VARCHAR(200);
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE customer ADD dod DATE DEFAULT '2021-02-02';
Query OK, 0 rows affected (0.10 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE customer DROP location;
Query OK, 0 rows affected (4.43 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE customer RENAME COLUMN cus_name TO name;
Query OK, 0 rows affected (7.50 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE customer MODIFY name VARCHAR(30);
Query OK, 0 rows affected (0.14 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> RENAME TABLE customer TO customer_info;
Query OK, 0 rows affected (0.02 sec)

mysql> TRUNCATE TABLE customer_info;
```

```
mysql> DROP TABLE customer_info;
Query OK, 0 rows affected (0.02 sec)

mysql> DESC customer_info;
ERROR 1146 (42S02): Table 'operations_db.customer_info' doesn't exist
mysql> CREATE TABLE persons (
    ->     id INT NULL,
    ->     lastname VARCHAR(255) NULL,
    ->     firstname VARCHAR(255) NULL,
    ->     age INT
    -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> DESC customer_info;
ERROR 1146 (42S02): Table 'operations_db.customer_info' doesn't exist
mysql> CREATE TABLE persons (
    ->     id INT NULL,
    ->     lastname VARCHAR(255) NULL,
    ->     firstname VARCHAR(255) NULL,
    ->     age INT
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE persons_nn (
    ->     id INT NOT NULL,
    ->     lastname VARCHAR(255) NOT NULL,
    ->     firstname VARCHAR(255) NOT NULL,
    ->     age INT
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE persons_check (
    ->     id INT,
    ->     age INT CHECK (age >= 18)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE persons_unique (
    ->     id INT UNIQUE,
    ->     lastname VARCHAR(255)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> CREATE TABLE persons_pk (
    ->     id INT PRIMARY KEY,
    ->     lastname VARCHAR(255)
    -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> CREATE TABLE orders (
    ->     order_id INT PRIMARY KEY,
    ->     order_number INT,
    ->     person_id INT,
    ->     FOREIGN KEY (person_id) REFERENCES persons_pk(id)
    -> );
Query OK, 0 rows affected (0.18 sec)

mysql> CREATE TABLE persons_default (
    ->     id INT,
    ->     city VARCHAR(100) DEFAULT 'Sandnes'
    -> );
Query OK, 0 rows affected (0.07 sec)
```

.

TASK-2

```
mysql> CREATE TABLE orders (
    ->     order_id INT PRIMARY KEY,
    ->     order_number INT,
    ->     person_id INT,
    ->     FOREIGN KEY (person_id) REFERENCES persons_pk(id)
    -> );
Query OK, 0 rows affected (0.18 sec)

mysql> CREATE TABLE persons_default (
    ->     id INT,
    ->     city VARCHAR(100) DEFAULT 'Sandnes'
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> INSERT INTO persons_pk VALUES (1,'Alex');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO persons_pk VALUES (2,'John');
Query OK, 1 row affected (0.01 sec)

mysql> UPDATE persons_pk SET lastname='David' WHERE id=2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> DELETE FROM persons_pk WHERE id=2;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT COUNT(*) FROM persons_pk;
+----------+
| COUNT(*) |
+----------+
|        1 |
+----------+
1 row in set (0.01 sec)
```

```
mysql> SELECT AVG(id) FROM persons_pk;
+---------+
| AVG(id) |
+---------+
|  1.0000 |
+---------+
1 row in set (0.00 sec)

mysql> SELECT * FROM persons_pk WHERE lastname LIKE 'A%';
+----+----------+
| id | lastname |
+----+----------+
|  1 | Alex     |
+----+----------+
1 row in set (0.01 sec)

mysql> SELECT DISTINCT lastname FROM persons_pk;
+----------+
| lastname |
+----------+
| Alex     |
+----------+
1 row in set (0.01 sec)

mysql> SELECT lastname, COUNT(*) FROM persons_pk GROUP BY lastname;
+----------+----------+
| lastname | COUNT(*) |
+----------+----------+
| Alex     |        1 |
+----------+----------+
1 row in set (0.00 sec)

mysql> SELECT * FROM persons_pk ORDER BY lastname;
+----+----------+
| id | lastname |
+----+----------+
|  1 | Alex     |
+----+----------+
1 row in set (0.00 sec)
```

TASK-3

```
mysql> SELECT DISTINCT p.id
    -> FROM persons_pk p
    -> INNER JOIN orders o
    -> ON p.id = o.person_id;
Empty set (0.00 sec)

mysql> SELECT id FROM persons_pk
    -> WHERE id NOT IN (SELECT person_id FROM orders);
+----+
| id |
+----+
|  1 |
+----+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE show_persons()
    -> BEGIN
    ->     SELECT * FROM persons_pk;
    -> END;
    -> //
Query OK, 0 rows affected (0.02 sec)

mysql> DELIMITER ;
mysql> DELIMITER //
mysql> CREATE FUNCTION person_count()
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE total INT;
    ->     SELECT COUNT(*) INTO total FROM persons_pk;
    ->     RETURN total;
    -> END;
    -> //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM persons_pk
    -> WHERE id = (SELECT MAX(id) FROM persons_pk);
+----+----------+
| id | lastname |
+----+----------+
|  1 | Alex     |
+----+----------+
1 row in set (0.01 sec)

mysql> SELECT p.id, o.order_number
    -> FROM persons_pk p
    -> INNER JOIN orders o
    -> ON p.id = o.person_id;
Empty set (0.00 sec)

mysql> SELECT p.id, o.order_number
    -> FROM persons_pk p
    -> LEFT JOIN orders o
    -> ON p.id = o.person_id;
+----+--------------+
| id | order_number |
+----+--------------+
|  1 |         NULL |
+----+--------------+
1 row in set (0.00 sec)

mysql> SELECT p.id, o.order_number
    -> FROM persons_pk p
    -> RIGHT JOIN orders o
    -> ON p.id = o.person_id;
Empty set (0.00 sec)

mysql> SELECT * FROM persons_pk CROSS JOIN orders;
Empty set (0.00 sec)
```

```
mysql> SELECT id FROM persons_pk
    -> UNION
    -> SELECT person_id FROM orders;
+------+
| id   |
+------+
|    1 |
+------+
1 row in set (0.00 sec)

mysql> SELECT DISTINCT p.id
    -> FROM persons_pk p
    -> INNER JOIN orders o
    -> ON p.id = o.person_id;
Empty set (0.00 sec)

mysql> SELECT id FROM persons_pk
    -> WHERE id NOT IN (SELECT person_id FROM orders);
+----+
| id |
+----+
|  1 |
+----+
1 row in set (0.00 sec)

mysql> DELIMITER //
mysql> CREATE PROCEDURE show_persons()
    -> BEGIN
    ->     SELECT * FROM persons_pk;
    -> END;
    -> //
Query OK, 0 rows affected (0.02 sec)
```

Task -4

```
mysql> DELIMITER ;
mysql> DELIMITER //
mysql> CREATE FUNCTION person_count()
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE total INT;
    ->     SELECT COUNT(*) INTO total FROM persons_pk;
    ->     RETURN total;
    -> END;
    -> //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> DELIMITER //
mysql> CREATE PROCEDURE loop_example()
    -> BEGIN
    ->     DECLARE i INT DEFAULT 1;
    ->     WHILE i <= 5 DO
    ->         SELECT i;
    ->         SET i = i + 1;
    ->     END WHILE;
    -> END;
    -> //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> DELIMITER //
mysql> CREATE TRIGGER phone_check
    -> BEFORE INSERT ON customer_info
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.contact NOT REGEXP '^[7-9]' THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Enter correct phone number';
    ->     END IF;
    -> END;
    -> //
```

```
mysql> SHOW TABLES;
+-------------------------+
| Tables_in_operations_db |
+-------------------------+
| audit_logs              |
| orders                  |
| persons                 |
| persons_check           |
| persons_default         |
| persons_nn              |
| persons_pk              |
| persons_unique          |
| tasks                   |
| users                   |
+-------------------------+
10 rows in set (0.00 sec)

mysql> RENAME TABLE customer TO customer_info;
ERROR 1146 (42S02): Table 'operations_db.customer' doesn't exist
mysql> DESC users;
+------------+----------------------------------+------+-----+-------------------+-------------------+
| Field      | Type                             | Null | Key | Default           | Extra             |
+------------+----------------------------------+------+-----+-------------------+-------------------+
| user_id    | int                              | NO   | PRI | NULL              | auto_increment    |
| name       | varchar(50)                      | NO   |     | NULL              |                   |
| email      | varchar(100)                     | YES  | UNI | NULL              |                   |
| password   | varchar(200)                     | NO   |     | NULL              |                   |
| role       | enum('Admin','Manager','Employee') | NO   |     | NULL              |                   |
| created_at | datetime                         | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------+----------------------------------+------+-----+-------------------+-------------------+
6 rows in set (0.00 sec)

mysql> ALTER TABLE users
    -> ADD contact BIGINT;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Task 05

```
mysql> DESC users;
+------------+----------------------------------+------+-----+-------------------+-------------------+
| Field      | Type                             | Null | Key | Default           | Extra             |
+------------+----------------------------------+------+-----+-------------------+-------------------+
| user_id    | int                              | NO   | PRI | NULL              | auto_increment    |
| name       | varchar(50)                      | NO   |     | NULL              |                   |
| email      | varchar(100)                     | YES  | UNI | NULL              |                   |
| password   | varchar(200)                     | NO   |     | NULL              |                   |
| role       | enum('Admin','Manager','Employee') | NO   |     | NULL              |                   |
| created_at | datetime                         | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| contact    | bigint                           | YES  |     | NULL              |                   |
+------------+----------------------------------+------+-----+-------------------+-------------------+
7 rows in set (0.00 sec)

mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER phone_check
    -> BEFORE INSERT ON users
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.contact NOT REGEXP '^[7-9]' THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Enter correct phone number';
    ->     END IF;
    -> END;
    -> //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> INSERT INTO users (name,email,password,role,contact)
    -> VALUES ('Test','test@mail.com','123','Employee',9876543210);
Query OK, 1 row affected (0.02 sec)

mysql> CREATE VIEW customer_view AS
    -> SELECT name, reg_id FROM customer_info;
```

```
mysql> CREATE PROCEDURE otp_check(IN otp INT)
    -> BEGIN
    ->     IF LENGTH(otp) <> 4 THEN
    ->         SIGNAL SQLSTATE '45000'
    ->         SET MESSAGE_TEXT = 'Enter 4 digit codes';
    ->     END IF;
    -> END;
    -> //
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql>
```